

3.3.

(1) 三种方法:

① 预测分支失败

② 预测分支成功

③ 延迟分支

(2) 若同是: 对分支处理方法
在程序执行过程中不变,
是静态的.

3.6 令 Δt 为 5ns

$$(1) TP = \frac{n}{\sum_i \Delta t_i + (n-1) \max \Delta t_i}$$

$$= \frac{10}{8\Delta t + 9 \times 4\Delta t}$$

$$= \frac{10}{44\Delta t}$$

$$= \frac{5}{22\Delta t} = \frac{1}{220} (\text{ns})^{-1}$$

$$\text{效率: } E = \frac{S}{K} = \frac{10 \times 8\Delta t}{4 \times (8\Delta t + 9 \times 4\Delta t)}$$

$$= \frac{80\Delta t}{4 \times 44}$$

$$= \frac{5}{11}$$

(2) 第4段

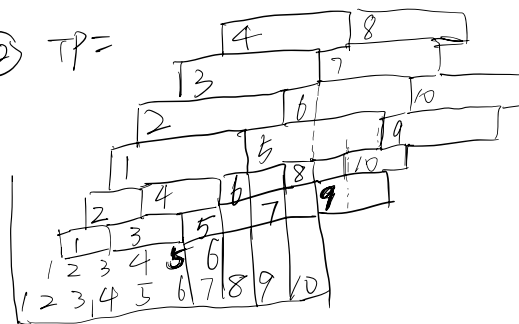
① 分段变成每段 5ns

② 3个并行预段并行

$$(1) TP = \frac{10}{400 + 9 \times 50} = \frac{1}{85} (\text{ns})^{-1}$$

$$E = \frac{10 \times 8\Delta t}{8(8\Delta t + 9\Delta t)} = \frac{10}{17}$$

(2) $TP =$



$$TP = \frac{10}{17\Delta t} = \frac{1}{85} (\text{ns})^{-1}$$

$$E = \frac{10}{17}$$

3.7. (1) 在第3段时发生冲突, 任务被阻塞
∴ 会优化成 每隔 $2\Delta t$ 启动一任务

$$(2) TP_{\max} = \frac{1}{2\Delta t}$$

$$TP = \frac{10}{5\Delta t + 9 \times 2\Delta t} = \frac{10}{23\Delta t}$$

$$E = \frac{10 \times 5\Delta t}{4(23\Delta t)} = \frac{25}{46}$$

(3) 并行, 即第3段2个作业依次
进入, 并行.

则有

$$TP_{\max}' = \frac{1}{\Delta t}$$

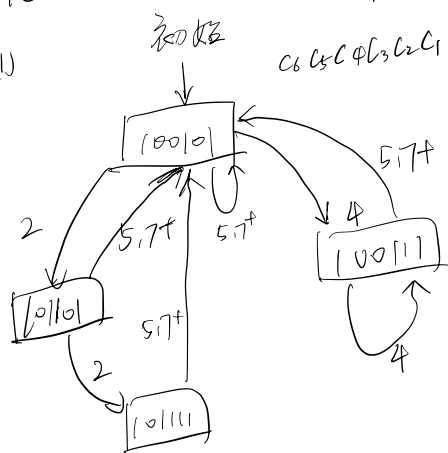
$$\text{此时 } TP' = \frac{10}{14\Delta t} = \frac{5}{7\Delta t}$$

3-10

(1)

$$F = \{1, 3, 6\}$$

$$C_0 C_1 C_2 C_3 C_4 C_5 \therefore C_0 = 10010$$



(2) ① 非等: $2 \rightarrow 2 \rightarrow 5 \rightarrow 2 \rightarrow 2 \rightarrow 5 \dots$

$$\therefore \text{平均间隔为 } \frac{2+2+5}{3} = 3$$

$$\therefore TP_{max} = \frac{1}{3\Delta t}$$

② 等间隔: $4, 4, 4, 4 \dots$

$$TP_{max} = \frac{1}{4\Delta t}$$

(3) ① 非等间隔: $TP = \frac{10}{7\Delta t + 27\Delta t} = \frac{5}{17\Delta t}$

$$\text{加速比: } S = \frac{10 \times 7\Delta t}{34\Delta t} = \frac{35}{17} \approx 2.06$$

② 等间隔: $TP = \frac{10}{7\Delta t + 36\Delta t} = \frac{10}{43\Delta t}$

$$S = \frac{10 \times 7\Delta t}{43\Delta t} = \frac{40}{43} \approx 1.63$$

3-12.

理想情况 $CPI = 1$

$$\text{时钟频率} + \text{周期}: 0.03 \times 1 = 0.05$$

分支 20%, 60% 成功,

$$\therefore 0.20 \times 0.60 \times 2 = 0.24$$

$$\therefore CPI = 1 + 0.05 + 0.24 = 1.29$$

$$\therefore \text{加速比} = \frac{1.29}{1} = 1.29$$